Schizophrenia recognition based on gramian angular field combining activation features: a functional near-infrared spectroscopy study

Mingxi Yang, Yulu Yang, Meiyun Xia, Huiting Qiao, Daifa Wang, Yajing Meng, and Di Wu

***Abstract*—In recent years, the incidence of schizophrenia has been increasing globally. Combination of functional near-infrared spectroscopy with verbal fluency task (fNIRS-VFT) provides an objective neurofunctional assessment tool for psychiatrists in auxiliary diagnosis. However, current methods face challenges such as insufficient time series feature extracting, inadequate feature utilization, and unstable robustness of single or few features. In this study we designed a fNIRS classification strategy by gramian angular field (GAF) coding integrating activation information. This strategy utilizes GAF normalized by the global maximum and minimum to via fNIRS signals into 2D representations virtual images for SCZ recognition. Specifically, fNIRS data of 200 participants were constructed and processed into virtual images. Then, the classification performance of five different processing methods, including directly using activation sequence, recurrence plot, markov transition field, GAF image without activation, and traditional fNIRS features is compared. Finally, this study additionally collected 40 cases of fNIRS-VFT data as an external test set. Compared with EfftivenetV2 with GAF images without activate information (73.4% accuracy) and CNN HbO signals classification (75.5% accuracy), the SCZ recognition accuracy based on GAF images combining activation information improved by 7.6% and 4.9%. The ShuffleNetV2 achieved the best performance with an accuracy of 81.0% on the cross-validation dataset, and obtained an accuracy of 72.0% on the external test set. Our findings indicate that GAF virtual image coding approach integrating activation information forms a new strategy for supporting SCZ screening and diagnosis. It further promotes the application of fNIRS technology in the field of clinical psychiatric disorders.**

***Index Terms*—functional near-infrared spectroscopy, schizophrenia, verbal fluency task, gramian angular field coding.**

I. INTRODUCTION

SCHIZOPHRENIA (SCZ) is a severe, chronic, and highly disabling psychiatric disorder [1], [2]. Patients with SCZ typically exhibit abnormal symptoms such as delusions, hallucinations [3], inadequate thinking [4], auditory hallucinations [5], [6], and impaired cognitive function[7] (attention [8], phonological deficits [9], [10], memory [4]). These symptoms significantly impact the patient's learning and life, and are a potential safety hazard for the patient and those around them. Currently, the diagnosis of SCZ relies on psychiatrists using the Diagnostic and Statistical Manual of Mental Disorders (DSM) in conjunction with symptom descriptions and behavioral observations of the patient [11], [12]. However, this approach not only considers the clinical experience of the psychiatrist, but also depends on the patient's cooperation. In recent years, evolving neuroimaging technology has provided psychiatrists with objective neuroimaging information about patients, which is expected to overcome the limitations of relying solely on clinical interviews to diagnose SCZ.

Brain functional imaging techniques, such as functional near-infrared spectroscopy (fNIRS), electroencephalography, functional magnetic resonance imaging, etc., provide effective and convenient tools for exploring brain function and measuring neural activity. Among them, fNIRS infers the relative changes in hemoglobin (Hb) in the cerebral cortex are inferred by measuring the intensity differences of specific wavelengths of near-infrared light (650-900 nm) passing through the brain. Based on neuro-vascular coupling [13], changes of Hb concentrations have been used as indicators of cerebral cortex activity. Compared with some classical brain imaging techniques such as fMRI, EEG or PET, fNIRS has received attention in the field of psychology and psychiatric disorders due to its low-cost, non-invasive, non-radiation, and patient-friendly properties [14], [15], [16], [17], [18], [19]. For example, in the clinical screening and dynamic evaluation of psychiatric disorders, fNIRS has higher operability and clinical feasibility compared to other neuroimaging tools. It is suitable for practical applications such as rapid brain function assessments that require minor constraints and high adaptability to different environments.

The verbal fluency task (VFT) is a cognitive task paradigm widely used in neuropsychological or neuroimaging testing [19].

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This task is easy to administer and requires a small amount of time and space to complete, which made it commonly used for psychiatric differential diagnosis in natural clinical environments [20], [21]. SCZ patients tend to have lower activation strengths in the prefrontal cortex (PFC) and superior temporal cortex (STC) than healthy subjects during the VFT task [2], [15], [22], [23]. The strength of functional connectivity (FC) in the PFC and STC was lower than that in the healthy group [2]. These abnormal cortical activities are used for brain function analysis and classification. Takizawa et al. in 2014 first developed two metrics, integral value (IV) and centroid value (CV), for identification of major psychiatric disorders. The accuracy in distinguishing between emotional disorders and SCZ patients reached 74.6% [15]. Wei et al. tested the composite index of IV and CV in 2021 and showed an AUC of 0.737 for distinguishing SCZ and healthy control (HC) [16]. Tran et al. classified HC and SCZ using IV of the right orbitofrontal cortex under the VFT task in 2023with an accuracy of 76% [24]. Although these biomarkers have been validated in studies, certain limitations remain. For example, subjects were included when the number of positively activated channels in frontal and temporal brain regions was greater than 50% [15], [16]. However, SCZ patients are often reported to have significant decreased activation [19] or no increase in HbO concentration in most channels [25]. This resulted in SCZ patients who did not show activation being excluded from those studies. In addition, when fNIRS-VFT technology is applied to complex clinical environments, the robustness of single feature classification is often unstable. Neurological markers or strategies to recognition for SCZ still need to be further explored.

Except the machine learning classifiers that rely on hand-crafted features, deep learning can achieve superior classification performance due to its powerful representation ability. Some researchers have used physiological signal time series as input for deep learning classifiers and achieved exciting results [26], [27], [28], [29]. Qian et al. designed a specific decoding structure for fNIRS data analysis based on CNN in 2021 to achieve high classification accuracy [30]. Ma et al. used CNN to classify fNIRS data of motor imagery in 2021, and achieved the best accuracy by single channel of 80% [29]. Although these classification methods have made some progress, they may not be ideal strategies for deep learning models [31]. With the deepening of research, the technology of converting time series into virtual images has been developed. Researchers have combined these virtual images with deep learning techniques and applied them to fields such as chemistry [32], optics [33], and physiological signal analysis [34], [35], [36].

Inspired by the superior results of encoding time series into images for classification in those researches, we consider whether this approach can provide a unique strategy for fNIRS-VFT SCZ classification research. SCZ patients have been reported to exhibit specific fluctuations in HbO concentration changes during VFT tasks. For example, the concentration of HbO in the cerebral cortex of SCZ patients showed a more gradual increase during the task, followed by inefficient HbO growth after the task was completed [37]. Therefore, compared to extracting fixed types of features, encoding time series into images is more effective in capturing subtle brain activity features in fNIRS signals. In addition, considering that virtual image encoding technology may eliminate information such as activation intensity of different groups. For this obstacle, we propose a GAF virtual image encoding strategy integrating activation information. Firstly, we collected HC group and SCZ group fNIRS-VFT data, then preprocessed the raw data. In this step, the raw data is converted into hemoglobin data. Subsequently, the GAF encoding technique utilizing the maximum and minimum values of global activation is used to convert the multi-channel HbO activation into a multi-channel virtual image. Third, the differences in activation intensity during the VFT task between HC and SCZ is compared. Fourth, we conducted four types of classification comparison experiments: comparison with HbO time series classification; comparison with other time series-to-image conversion methods (such as the recurrence plot (RP) [38] and markov transition field (MTF) [39] methods); and comparison with GAF virtual images without activation intensity features. At the same time, we compared the classification performance of the GAF virtual image-based method with that of traditional features commonly used in fNIRS studies. Finally, an additional dataset was collected in this study to serve as an independent test set for generalization validation. According to our knowledge, this is the first study to use combining activation intensity GAF image encoding technology to process and analyze fNIRS-VFT for psychiatric disorders. We hope that the findings will inspire more fNIRS psychiatric disorders studies. The novelty and contributions of this paper include:

- We proposed an fNIRS-VFT classification strategy that generates multi-channel virtual images by combining activation features GAF encoding. These virtual images preserve characteristics such as activation intensity and rate, and can be directly used by deep learning models;
- We collected and constructed a dataset of 200 fNIRS-VFT recordings. Four different activation signal processing methods were examined for their impact on classification performance, including direct use of time series, RP, MTF, and GAF encoding without activation. In addition, commonly used traditional features in the fNIRS field were comprehensively compared. The comparison results show that the GAF image with fused activation features has the best classification effect;
- We additionally collected data from 40 participants as an external test set to verify the generalization ability of the proposed method. The best classification performance is achieved when GAF images combining activation intensity and a lightweight deep learning model. The classification probability may assist clinicians in evaluation and diagnosis.

The rest of this paper is organized as follows. Related works are briefly reviewed in Section II. Section III expounds the process of fNIRS scanning of different groups participants and data preprocessing. Section IV introduces the framework of GAF virtual image encoding combining activation features, and traditional fNIRS feature extraction methods. Section V describes experimental setup. Section VI reports the experimental results. Discussion, conclusion and future work are presented in Sections VII and VIII, respectively.

II. Relate Work

In this section we discuss related work. First, we discuss existing work on SCZ recognition based on fNIRS-VFT. And then we introduce our main method on virtual image generation of time series.

A. fNIRS Schizophrenia Recognition and Classification

At present, researchers often design and use different experimental paradigms based on their own needs in fNIRS research on SCZ, such as resting-state scans, memory tasks, speech tasks, emotional tasks, etc. Due to the short time consumption and ease of completion of VFT tasks, it is suitable for clinical examinations. Besides, different experimental paradigms have different evaluation criteria. Therefore, in this paragraph, we discuss the classification of SCZ under the fNIRS-VFT, rather than resting state or other tasks. In addition to the IV and CV mentioned in the first section, statistical features such as mean, variance, peak, and slope of changes in blood oxygen concentration are also commonly used in machine learning classifiers such as Support Vector Machine (SVM), Linear Discriminant Analysis (LDA), and Random Forest (RF) [15], [40], [41], [42]. Zhang et al. used mean of activation changes under VFT in 2023 to distinguish adolescents with first-episode SCZ. The results showed that the AUC of concentration changes in the channels of the frontopolar area, dorsolateral prefrontal cortex, and the triangularis Broca's area was greater than 0.8, which could effectively recognize SCZ [7]. Chou et al. used the CV of average activation of the frontal lobe region in 2023 to identify different psychiatric disorders. The results showed that under the optimal threshold conditions (CV of 54 seconds), unipolar depression patients (72.5%), bipolar disorder (78.9%), and SCZ (84.6%) could be distinguished from HC [41]. Diao et al. identified SCZ from HC using activation characteristics (IV, CV, activation slope) of different channels in 2024. The accuracy of using decision tree classifier is the highest, at 69.2% [43].

In addition to the characteristics of changes in blood oxygen concentration signals, some researchers have used FC between different channels as a strategy for SCZ recognition. Yang et al. first used single channel FC intensity to identify SCZ under VFT in 2020. The highest accuracy of 84.67% was achieved by combining the FC of the three channels in the frontal lobe and the lateral layer of the left frontal lobe [3]. Ji et al. modeled and classified using FC formed by channels with significant activation differences between SCZ group and HC group under VFT in 2020. The average classification accuracy of different FC combinations is similar, ranging from 85.33% to 89.67% [44]. Eken et al. collected fNIRS data on HC, bipolar disorder, and SCZ under a speech task in 2022. Researchers classified different diseases by extracting the mean of dynamic functional connections. The classification results of 5-fold cross validation showed an average classification rate of 82.5% for distinguishing HC and SCZ, 75.5% for HC and bipolar disorder, and 79.0% for SCZ and bipolar disorder [17]. Current research lacks the consideration of combining both activation features and FC of different brain regions. Although these features were used as baseline comparisons in this study, we extracted as many features of all types as possible.

B. Encoding Time Series to Images

In recent years, a few emerging ways of mapping one-dimensional sequences to two-dimensional images have been developed, like RP [31], MTF, and GAF [39]. Among them, GAF coding is a transformation framework for mapping time series into gramian angular summation filed (GASF) images and gramian angular difference filed (GADF) images [32], [36]. One-dimensional time series data arranged by wavelength or channel are converted into multi-channel two-dimensional images by GAF encoding. In the field of chemistry, Jiang et al. designed a GASF deep learning classification model for monitoring aflatoxins in moldy peanuts. Compared with machine learning models, this approach improves monitoring performance and has stronger robustness [32]. Tan et al. used GAF encoding to analyze the infrared qualitative problem of microplastics, and the classification accuracy of using GASF and GADF images was about 13% higher than that based on traditional feature classification [45]. In the biomedical field, Ma et al. applied GAF encoding technology to identify different types of arrhythmias. The results showed that the method of converting electrocardiogram time series into GAF virtual images can characterize rich electrocardiogram features with the highest accuracy and specificity [36]. In the field of fNIRS, researchers attempted to use this method to classify fNIRS signals in resting and task states, and the results showed that GADF images performed better than fNIRS signal classification [31]. Zhang et al. used GAF encoding technology to convert one-dimensional fNIRS signals of HC and patients with mild cognitive impairment into two-dimensional images for auxiliary diagnosis. The results indicate that the classification performance using GAF virtual images is better to using fNIRS signal [34]. GAF image encoding technology can explore the potential waveform variation characteristics, correlation before and after different time points, consistency and other information in time series signals. This technology has broad application prospects in feature calibration and classification of time series [31], [32], [33], [36], [45]. However, we find that conventional GAF encoding may weaken differential brain activation information. In addition, since few studies apply GAF to fNIRS classification of mental disorders, we conduct more indepth studies.

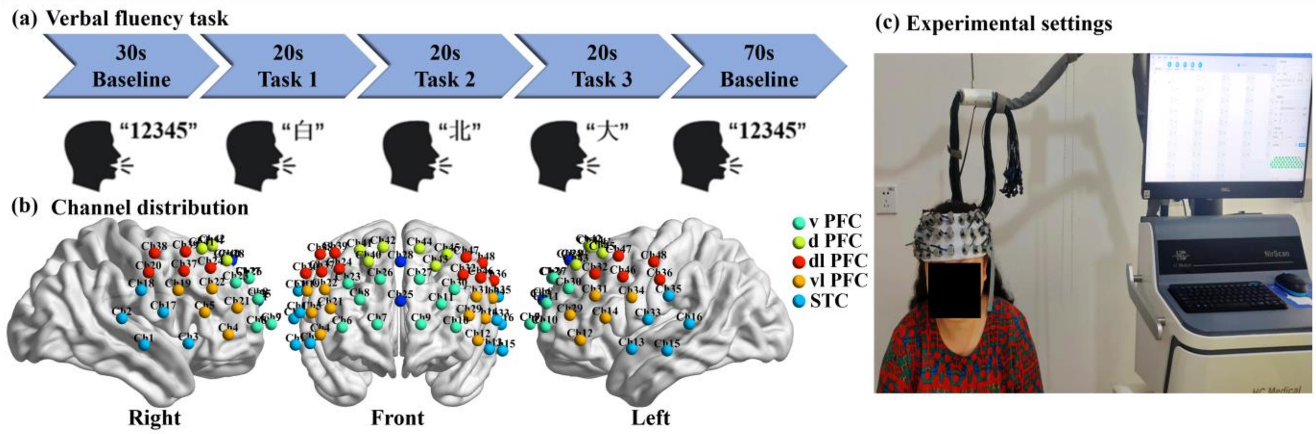

Fig. 1. (a) VFT process; (b) Channel distribution; (c) Experimental settings.

III. Data Preparation And Preprocessing

The dataset used in this paper was collected from the Mental Health Center of Sichuan University West China Hospital in Sichuan Province and the volunteers were tested in experimental conditions. This section presents the details of data collecting and preprocessing.

A. Participants

Outpatients with a clinical diagnosis of SCZ and HC were recruited by the Mental Health Center of Sichuan University West China Hospital in Sichuan Province. Each patient was diagnosed by trained psychiatrists using the Structured Clinical Interview for DSM-IV Axis I Disorders. Exclusion criteria included comorbid psychiatric disorders, alcohol or substance abuse, and patients with traumatic brain injury. This study was conducted in accordance with the Declaration of Helsinki and approved by the Biomedical Ethics Review Committee of West China Hospital, Sichuan University, China in August 2022 (ethics approval number: 2022(1282)). A total of 200 participants were recruited in this study, including 100 patients with SCZ and 100 HC. All participants were informed about the task procedures before the experiment. Before analyzing fNIRS data, we first excluded subjects who were either too young or too old (age <18 or age >45), and then excluded subjects with poor signal quality [46], [47] (defined as having more than 10% of channels with fNIRS signal coefficient variation > 10% of the total number of channels). A total of 184 samples were included in the final analysis, comprising 92 SCZ and 92 HC participants. Independent sample t-tests were conducted to assess age group differences (p = 0.831), and gender proportions were analyzed using the $\chi 2$ test (p = 0.238). The demographic information of the subjects was given in **Table I**.

TABLE I
Demographical Information of Participants.

	SCZ	HC	p-val
Age (years)	28.48±7.60	28.72±7.56	0.831
Gender (M/F)	52/40	44/48	0.238

B. Verbal Fluency Task

The VFT used in this study was similar to previous studies [15], [16]. The entire experimental flow is shown in Fig. 1(a). Each task consisted of a 30-second pre-task baseline scan, a 60-second task period subdivided into three 20-second blocks, and finally, a 70-second post-task baseline scan. The total duration of the VFT was approximately 3 minutes. During the baseline scan, participants were asked to repeatedly count from 1 to 5. During the task, subjects were instructed to generate as many words as possible starting with three Chinese syllables ("白," "北," and "大," meaning white, north, and big, respectively). The syllables changed each 20 second interval. Cues for syllable changes were given to subjects as audio feedback.

C. fNIRS Scan and Preprocess

1) fNIRS Scan: The fNIRS signals were acquired using a multi-channel, three-wavelength (730, 808, and 850 nm) functional near-infrared brain imaging device (NirScan-6000A, Danyang Huichuang Medical Equipment Co., Ltd, China) at a sampling rate of 11 Hz. The 15 light sources and 16 detectors form a total of 48 channels. The distance between the light sources and detectors is 3 cm. According to the international 10-20 system, the lowest light sources and detectors were placed along the T4-Fpz-T3 line, as shown in Fig. 1(b). The Montreal Neurological Institute (MNI) standard coordinates corresponding to each channel number are shown in Part 1 of supplementary material. This configuration allows scanning for changes in hemoglobin concentration in bilateral PFC and STC. All participants completed the fNIRS-VFT scan in a quiet room. During this time, they were required to focus their attention and minimize head movements. The process of data collection is shown in Fig. 1(c).

2) Data Preprocessing: Before data preprocessing, we checked data quality using coefficient variation [46], [47]. Data were excluded if the number of channels with coefficient variation > 10% accounted for more than 10% of all channels. Then Homer2 in Matlab was used for data preprocessing. The raw data is first transformed into optical density data. Motion artifacts were subsequently removed using a modified spline method (STDEVthresh = 8, AMPthresh = 0.5, TMotion = 0.5 s, TMASK = 1 s) [48]. A sliding window was used to calculate the local variance of the fNIRS signal, and segments exceeding a preset threshold were marked as motion artifacts. Second,

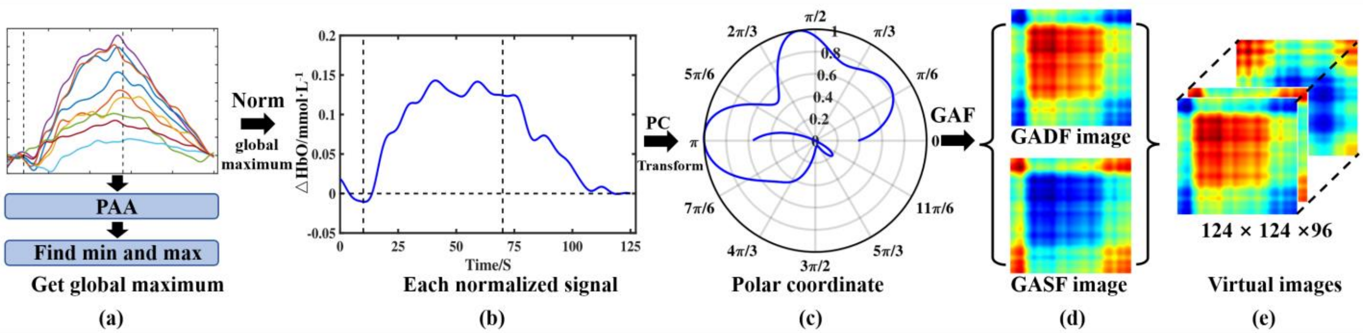

Fig. 2. GAF encoding workflow: the steps for converting HbO concentration changes into GAF virtual images combining activation features. (a) Apply the piecewise aggregation approximation (PAA) method to all HbO concentration change sequences to reduce the time series length (the data was shortened from 1375 points to 124 points), and determine the global minimum and maximum values based on all reduced sequences; (b) Display the HbO concentration variation curve normalized by the global minimum and maximum values for each channel; (c) Convert the normalized sequences from Cartesian coordinates to polar coordinate representation; (d) Generate combining activation intensity information GADF and GASF virtual images based on the polar coordinate angles; (e) Alternately stack the GADF and GASF images generated for each channel in sequence to construct the complete multi-channel virtual image set.

Spline interpolation was applied to the marked segments to correct baseline drift. Then, Residual spike noise in the corrected signal was re-detected and marked. The marked spike artifacts were replaced with Gaussian white noise of the same variance to prevent short-duration spikes from affecting subsequent analysis. A bandpass filter with cut-off frequencies of 0.00–0.10 Hz was used to remove physiological noise [25]. Finally, changes in Oxy-Hb (HbO) and Deoxy-Hb (HbR) were obtained from optical density data using the modified Beer-Lambert law (MBLL). Consistent with previous studies [15], [49], the HbO change curve during the task was obtained by linear fitting between a 10 s baseline before the task (the time period between 10 s and 0 s before task) and a 5 s baseline 50 s after the task (the time period between 50 s and 55 s after the task). Therefore, the duration of each HbO change curve is 125 s (10 s: pre-task period, 60 s: task period, 55 s: post-task period). In this study, we used HbO changes for subsequent analysis and feature extraction because HbO more directly reflects task-related cortical activation [15], [50].

In this study, the channels formed by the source-detector arrangement were divided into 10 regions of interest (ROI) based on their location, as shown in Fig. 1(b), including right ventral prefrontal cortex (R vPFC), left ventral prefrontal cortex (L vPFC), right dorsal prefrontal cortex (R dPFC), left dorsal prefrontal cortex (L dPFC), right ventrolateral prefrontal cortex (R vlPFC), left ventrolateral prefrontal cortex (L vlPFC), right dorsolateral prefrontal cortex (R dlPFC), left dorsolateral prefrontal cortex (L dlPFC), right superior temporal cortex (R STC), left superior temporal cortex (L STC). The HbO change curve for each ROI is the average of the HbO changes for all channels within that region.

IV. Methods

A. Combining Activation Intensity GAF Coding

GAF is a new method to encode a one-dimensional time series signal into a two-dimensional image representation without loss of features [31], [45]. The entire conversion process is shown in Fig. 2. For a time serie of size M × C HbO concentration variation, where M is the sampling point (125s × 11Hz) and C is the number of channels (48). We generate virtual images by following the steps below:

First, in order to reduce computational complexity [51], [52], the time series needs to be shortened to an appropriate length while maintaining its trend. Therefore, the length of the time series of HbO concentration changes was reduced (1375 to 124) by piecewise aggregation approximation (PAA) method, while preserving the time series fluctuation trend [31]. Then, the data for each channel was normalized between [-1, 1] by equation (1):

$$\tilde{x}_i = [(x_i - X_{max}) + (x_i - X_{min})]/(X_{max} - X_{min}) \quad (1)$$

Where x_i is the i-th sampling point for each channel, and $\tilde{x}_i$ is the normalized sampling point, $i = 1, 2, 3, \ldots, M$. In previous studies, X_{min} and X_{max} were the maximum or minimum values of each channel [31], [34]. However, this normalization strategy will first eliminate the activation differences of channels within different ROIs of the same subject. Secondly, the different activation intensity information between HC and SCZ groups will be lost. In this study, to avoid this problem, X_{min} and X_{max} are the maximum and minimum values of all subjects across all channels. Then, $\tilde{x}_i$ is transformed from the cartesian coordinates to the polar coordinate system by equation (2):

$$\begin{cases} \theta_i = \arccos(\tilde{x}_i), -1 \le \tilde{x}_i \le 1 \\ r_i = t_i/N \end{cases} \quad (2)$$

Where t_i is the timestamp, and N is a constant factor that normalizes the range of the polar coordinate system. This polar coordinate-based representation is a novel approach for understanding time series. As time progresses, the corresponding values are warped between different angular points on the spanning circle, like water rippling [39]. Finally, the trigonometric difference (GADF virtual image) and trigonometric sum (GASF virtual image) between each time point are calculated using equations (3) and (4), respectively, enabling easy extraction of temporal correlations between different time intervals.

Similar to previous research, the GADF and GASF generated from the HbO concentration change profiles of each channel

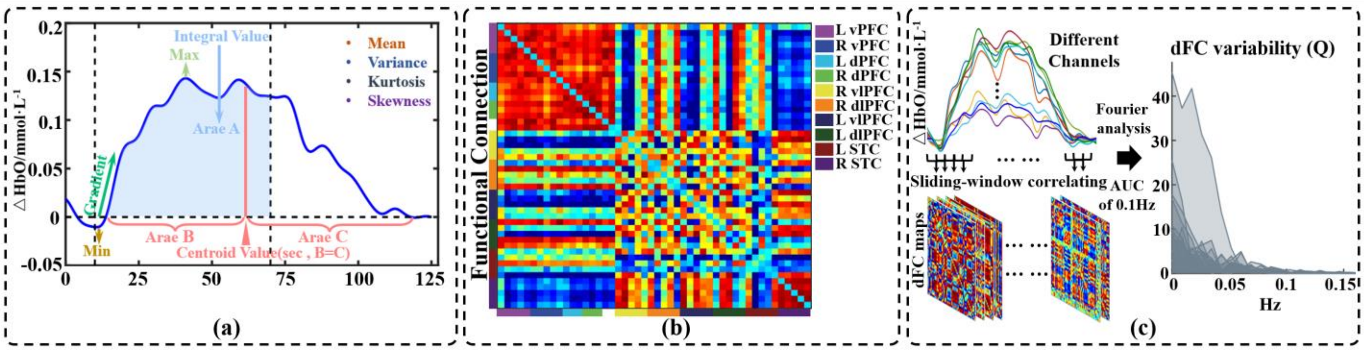

Fig. 3. Traditional fNIRS features, including (a) HbO activation curve features; (b) functional connectivity; (c) dynamic functional connectivity coefficient of variation.

were stored alternately to form the virtual images set [31], which used for the deep learning classifier. For each subject, the size of the virtual image set was 124 × 124 × 96.

$$GADF = \begin{bmatrix} sin(\theta_1 - \theta_1) \cdots \sin(\theta_1 - \theta_M) \\ \vdots \quad \ddots \quad \vdots \\ sin(\theta_1 - \theta_M) \cdots \sin(\theta_M - \theta_M) \end{bmatrix} \quad (3)$$

$$GASF = \begin{bmatrix} cos(\theta_1 + \theta_1) \cdots \cos(\theta_1 + \theta_M) \\ \vdots \quad \ddots \quad \vdots \\ cos(\theta_1 + \theta_M) \cdots \cos(\theta_M + \theta_M) \end{bmatrix} \quad (4)$$

B. Traditional fNIRS Feature Extraction

1) Activation Characteristics: Firstly, we calculated the time-series features of the activation curves (10 ROIs) for each brain region, including the mean of HbO concentration over 125 s, the variance of activation during the task period, the IV during the task period, the CV of the activation curves, the gradient of the activation change 5 s after the beginning of the task, maximum, minimum, kurtosis, and skewness. These characterizations are shown in Fig. 3(a). IV is the integral of the HbO concentration change curve during the task, representing the total magnitude of hemodynamic response in the subject's cerebral cortex during the VFT. CV is the time corresponding to the center of gravity of the area under the 125s HbO curve. A smaller CV indicates an earlier hemodynamic response and represents a more rapid cortical relaxation after the subject completes the VFT. Conversely, a larger CV indicates delayed cortical relaxation [15], [16], [19], [49]. The gradient of the activation change after the beginning of the task represents the rate at which the brain cortex responds after the participants begin to perform the VFT.

2) Functional Connection: In addition to the temporal characteristics of the HbO concentration change curves, we constructed FC between the 48 channels by Pearson correlation coefficients as follows:

$$r = \frac{\sum_{i=1}^{n}(A_i - \bar{A})(B_i - \bar{B})}{\sqrt{\sum_{i=1}^{n}(A_i - \bar{A})^2}\sqrt{\sum_{i=1}^{n}(B_i - \bar{B})^2}} \quad (5)$$

where A and B are the HbO activation curves for different channels, and r is the Pearson correlation coefficient. Thus, we obtained a 48 × 48 FC matrix for each participant. These correlation coefficients were subsequently converted to z-scores using Fisher's r-to-z transformation to improve normality. Finally, we averaged the FC matrices formed between channels within different ROIs to obtain 45 FC strengths, as shown in Fig. 3(b).

3) Dynamic Functional Connection Variability Index: In this study, we also focused on the stability of each participant's dFC. Firstly, we obtained the dFC time series for each participant using the sliding window correlation (SWC) approach. The process is shown in Fig. 3(c). Specifically, for each time series of channels, a 10-second time window was selected and moved in 1-second increments along the entire time length. Simultaneously, using equation (5), the FC between channels within each time window was calculated. The 116 sliding time windows are formed by 125 seconds of total time length and 10-second windows. Thus, 116 dFC time series were constructed. Subsequently, we calculated the dFC variability index Q for each participant [53], [54]. For the dFC sequence, we computed its spectral power using the Fourier transform. The dFC variability index (Q) is the area under the curve of the power spectrum in the low-frequency (< 0.1 Hz) band [53], [54]. Finally, we averaged the Q formed between channels within different ROIs to obtain 45 Q strengths. Larger Q values indicate that the dFC is more variable and unstable throughout the task, while smaller Q values indicate that the dFC is less variable and relatively stable [53], [54].

V. Experiments

In this study, we focused on evaluating the effectiveness of using GAF virtual images incorporating activation intensity to distinguish SCZ, which were transformed from fNIRS-VFT data.

First, we compared the activation intensity differences across various brain regions between the two groups by t-test to demonstrate the significance of activation intensity in SCZ identification.

Second, to verify the necessity of the GAF-based classification method, we conducted three comparative ablation experiments: (1) Comparing the classification performance of virtual GAF images with that of deep learning models directly applied to HbO activation time series; (2) Comparing the classification performance of GAF virtual images with other time-series-to-image conversion methods, including RP and

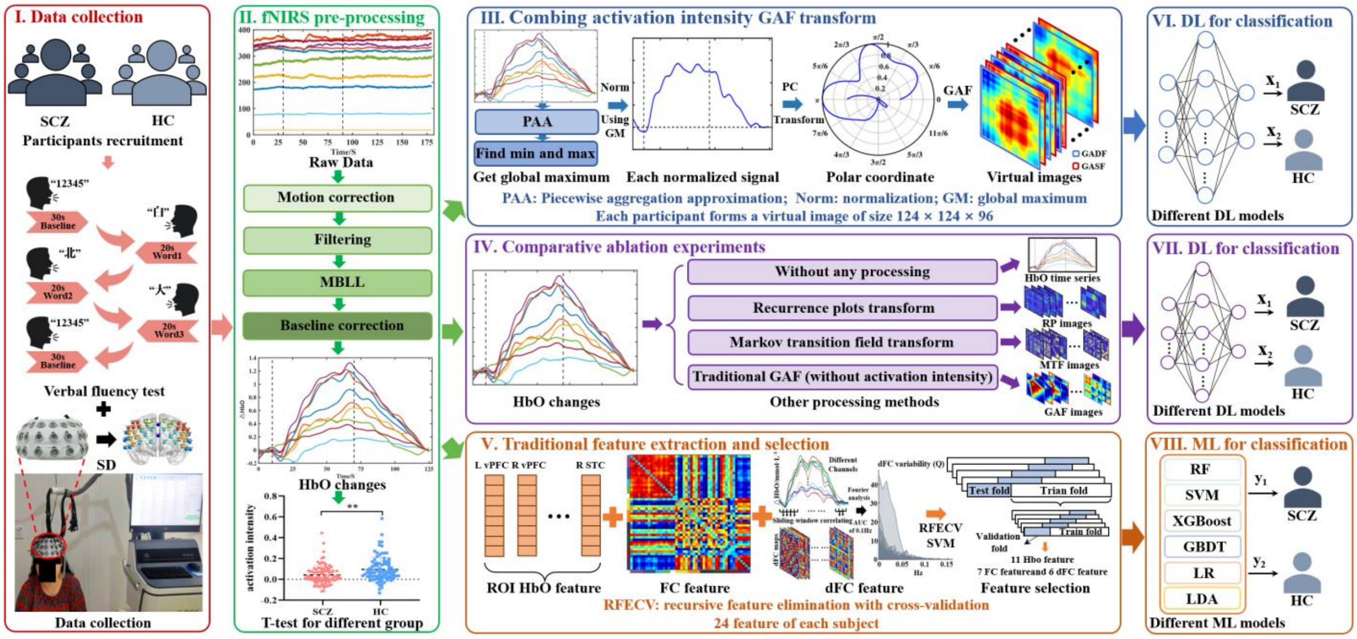

Fig. 4. The overview of method this study, including fNIRS scan, data preprocessing, GAF combining activate information virtual image encoding, other virtual image encoding, fNIRS feature extraction, feature selection, and model training.

MTF, to demonstrate the superiority of using GAF virtual images in fNIRS signal classification; (3) Comparing GAF virtual images containing activation intensity with traditional GAF images to validate that incorporating activation intensity into GAF encoding positively contributes to model discriminability and enhances the sensitivity of SCZ identification.

Third, we compared the classification performance of activation-intensity-based GAF virtual images with traditional features commonly used in the fNIRS, aiming to demonstrate the effectiveness and advantages of our GAF-based encoding method in extracting spatiotemporal features and improving classification performance.

Building on this foundation, an additional 40 subjects (20 HC and 20 SCZ) were collected as an independent external validation dataset to conduct classification testing, aiming to evaluate the generalization capability of the proposed GAF-based virtual image encoding method in terms of classification performance. The overall workflow of the study is illustrated in Fig. 4.

A. Experiment Setup

1) Deep Learning for Classification: Considering the size and amount of data used in the study, CNN, CNN-3b, RNN, LSTMs, Time Series Transformer (Ts-Transformer), Simple Time Series Transformer (Sts-Transformer) were used to classify HbO time series. ShuffleNetV2 [55], MobileNetV2 [56], EfficientNetV2 [57], ResNet18, ViT, ResMLP, PVTv2-B0, and MLP-Mixer [31] were used to classify all virtual images data. Refer to previous research, the models were all configured to have a batch size of 16 to ensure a balance of efficiency and performance during training. The models were trained for 150 epochs [31]. Iterations were performed using a cosine learning rate. Overfitting can degrade the classification performance and generalization ability of the model due to the limited amount of fNIRS data. Therefore, our study adopts a simple and efficient flooding regularization method, which is consistent with a previous study [31]. This regularization limits the training loss to a smaller constant value around zero. This value is 0.1 in this study. Virtual images were trained and classified using Python 3.8.16 and trained on NVIDIA GeForce RTX 3090 GPUs.

2) Machine Learning for Traditional Features Classification: Too many features are highly likely to cause overfitting of the model during classification. Recursive Feature Elimination and Cross Validation (RFECV) [58] were used to select the best features among different groups before classification. In order to find influential features among multiple features, the importance of each feature was obtained by RFE. Then, the best features are identified through cross-validation to obtain optimal classification performance. In our study, 24 features (11 HbO curve features, 7 FC features, 6 dFC features) were selected by the RFECV method for machine learning modeling. Given the limited number of data samples, six supervised classifiers (SVM, Logistic Regression (LR), RF, LDA [15], [40], [41], [42], XGboost [59], and Gradient Boosting Decision Tree [60] (GBDT)) were used to model and classify the relationships between selected features and diagnostic labels.

3) Evaluation Indicators: For both deep learning classifiers and machine learning classifiers, we used 5-fold cross-validation conducted across the entire dataset to evaluate the performance of the different approaches (All samples were randomly divided into five folds. In each iteration, four folds were used for training and one fold for validation, and the process was repeated five times, ensuring a balanced class distribution across the training and validation sets. The individuals in the training and testing sets used for deep learning classifiers and machine learning classifiers are consistent). Accuracy (Acc), Precision (Pre), Recall, and F1 scores were used to quantify the performance of these approaches for classification between groups.

VI. Result

A. Cortical Activation Among Different Groups

During the VFT task, the HC group showed higher average activation on vPFC, vlPFC, and STC compared to the SCZ group. The summary of the average activation comparisons across different brain regions is presented in Table II. 48 channels are mapped onto the cerebral cortex, as shown in Fig. 5. The activation pattern of the SCZ group in vPFC and STC is not obvious.

In the subsequent classification analysis, the GAF transformation was performed based on channel-wise activation curves. Therefore, we also conducted a physiological analysis of cortical activation across different channels between the HC and SCZ groups. The results of this analysis are presented in Part 2 of the supplementary materials.

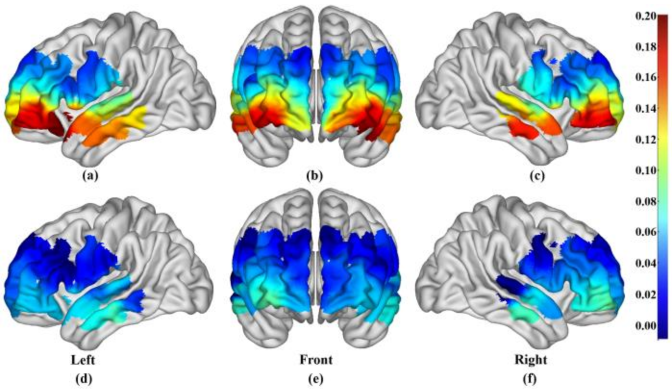

Fig. 5. The average activation topographic map of the cerebral cortex during VFT tasks in different groups. (a-c) are the left view, front view, and right view of the HC group, respectively; (d-f) are the left view, front view, and right view of the SCZ group, respectively.

TABLE II
Activation differences between the SCZ and HC groups

ROI	HC	SCZ	*p*-val (t-test)
Lef_vPFC	0.110 ± 0.018	0.059 ± 0.014	0.040
Rig_vPFC	0.119 ± 0.018	0.053 ± 0.015	0.012
Lef_dPFC	0.042 ± 0.011	0.021 ± 0.010	0.164
Rig_dPFC	0.037 ± 0.011	0.033 ± 0.011	0.766
Lef_vlPFC	0.118 ± 0.017	0.051 ± 0.014	0.003
Rig_vlPFC	0.098 ± 0.014	0.070 ± 0.013	0.110
Lef_dlPFC	0.050 ± 0.012	0.020 ± 0.011	0.238
Rig_dlPFC	0.048 ± 0.011	0.033 ± 0.010	0.685
Lef_STC	0.121 ± 0.019	0.062 ± 0.017	0.023
Rig_STC	0.122 ± 0.017	0.046 ± 0.018	0.002

B. Combining Activation Intensity GAF Virtual Images

The HbO concentration changes in all channels of all participants were converted into GAF virtual images for use in the deep learning classifier. Considering that executing VFT is associated with prefrontal cortex activity, parts (a) and (b) of Fig. 6 shows an example of HbO concentration change curves of channel 6 (located in the prefrontal cortex) and the corresponding GADF and GASF virtual images for different participants during the VFT task. Different groups correspond to different colors of HbO concentration change curves (HC in blue, and SCZ in red). The fluctuations of the HbO concentration change curve in HC, such as the wave peaks, correspond to the more prominently colored regions in the visualised GADF and GASF virtual images (circular markers and square markers in the parts (c) and (d) of Fig. 6). For SCZ patients, the inefficient growth of HbO after the VFT task is reflected in the square marked position in the second column of row (d) in Fig. 6.

In addition, considering the intensity of the HbO curves, we chose the maximum and minimum values of all the data for normalization. In this way, the intensity information is expressed as color shades in the visualised GADF and GASF virtual images. In Fig. 6, parts (c) and (d) represent the GAF images combining activation intensity. On the contrary, (e) and (f) represent ordinary GAF images. By comparison, we can observe that conventional GAF encoding loses activation strength information.

C. Comparison With Other Classification Method

1) Classification based on time series: The first part of Table III presents the results of SCZ identification using HbO time series. In this section, CNN-based architectures outperformed RNNs, LSTMs, and Ts-Transformer (CNN achieved an accuracy of 75.5% in 5-fold cross-validation and 61.2% on the external test set). Although RNNs are considered suitable for modeling time series, similar to previous studies, RNNs performed poorly in this study [31]. Hemodynamic responses may show no significant activation differences in certain brain regions, making it difficult for RNNs to capture discriminative information for classification. The Sts-Transformer and Ts-Transformer models achieved better performance than recurrent models on certain evaluation metrics. However, their overall performance was still inferior to that of CNN.

2) Classification based on Virtual Image: We further compared the effectiveness of other time series processing methods, including no transformation, RP, MTF, and GAF without activation features. The results of this part are summarized in the second section of Table III.

Different processing methods led to performance differences. On the 5-fold cross-validation dataset, the GAF images combining activation features achieved the best performance. ShuffleNetV2 obtained the highest classification accuracy under this encoding method (80.4% ± 7.4%). In contrast, GAF images without activation features performed slightly worse across all models. RP and MTF images showed overall lower performance, with the highest classification accuracy for RP images being 72.8% ± 5.8%, and for MTF images being 70.7% ± 3.2%, both lower than that of the GAF images with activation features.

On the external test set, the GAF images with activation features also showed relatively strong identification ability. Under this encoding method, the highest accuracy (72.0% ± 1.7%) was achieved, outperforming the GAF without activation features (67.1% ± 3.7%), RP (65.4% ± 3.9%), and MTF images (66.8% ± 2.7%).

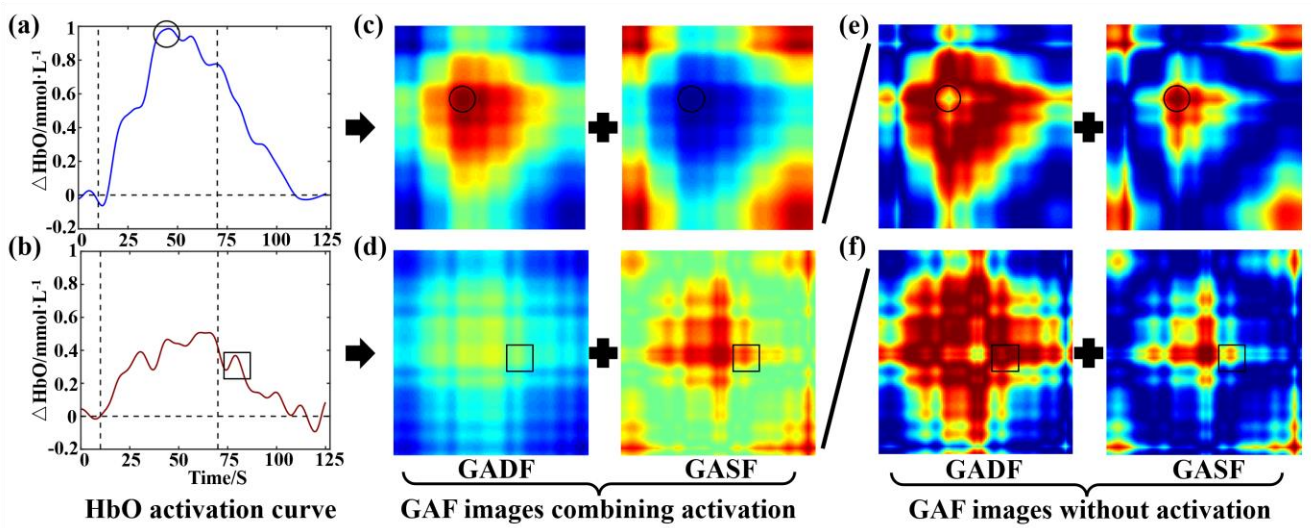

Fig. 6. (a) and (b) is an example of the HbO concentration change curve in channel 6 (prefrontal cortex): HC is blue and SCZ is red. (c) and (d) are the GADF and GASF virtual images of fusion activation features for HC and SCZ patients, respectively; (e) and (f) are GADF and GASF generated by traditional GAF encoding, which lose the activation strength feature. Among them, the highest activation point (circular area) on the HbO curve in Figure (a) is displayed as the brightest spot in the virtual image; (b) The fluctuations (square area) in the HbO curve are reflected as fluctuating color clusters in the virtual image.

TABLE III CLASSIFICATION PERFORMANCE OF SCZ AND HC UNDER DIFFERENT TIME SERIES PROCESSING METHODS

Data	Input	Classifier	Accuracy	Precision	Recall	F1	Data	Input	Classifier	Accuracy	Precision	Recall	F1
5-fold CV data set	HbO time series	CNN	75.5 ± 5.6	75.8 ± 5.3	75.1 ± 5.4	75.1 ± 5.6	external test set	HbO time series	CNN	61.2 ± 1.3	62.0 ± 1.0	61.7 ± 1.1	61.4 ± 1.2
		CNN-3b	72.3 ± 4.0	73.7 ± 4.5	71.9 ± 4.3	71.5 ± 4.2			CNN-3b	61.6 ± 2.0	61.7 ± 2.1	61.6 ± 2.0	61.5 ± 2.0
		LSTMs	62.5 ± 4.9	63.8 ± 3.7	62.0 ± 4.4	60.5 ± 5.9			LSTMs	50.4 ± 2.2	50.2 ± 3.0	50.4 ± 2.2	48.0 ± 4.0
		RNN	65.8 ± 4.2	67.4 ± 5.8	65.0 ± 3.7	64.4 ± 3.6			RNN	51.7 ± 4.3	51.7 ± 4.5	51.7 ± 4.6	51.2 ± 4.8
		Sts-Trans	62.5 ± 2.7	62.8 ± 2.4	69.7 ± 5.7	61.9 ± 3.2			Sts-Trans	54.5 ± 5.3	54.7 ± 5.6	62.0 ± 8.7	54.1 ± 5.4
		Ts-Trans	65.7 ± 7.0	67.0 ± 7.1	71.0 ± 10.5	64.4 ± 8.6			Ts-Trans	58.0 ± 2.9	59.1 ± 3.6	69.0 ± 12.4	56.9 ± 3.0
	GAF images with HbO Int	ReNet18	77.7 ± 5.4	79.7 ± 4.7	77.4 ± 5.2	77.0 ± 5.9		GAF images with HbO Int	ReNet18	71.7 ± 3.5	72.0 ± 3.3	65.7 ± 6.0	71.5 ± 3.6
		MoNetV2	73.9 ± 7.3	74.6 ± 7.7	73.7 ± 8.0	73.4 ± 7.6			MoNetV2	64.3 ± 1.5	64.5 ± 1.6	64.7 ±2.7	64.3 ±1.5
		ShuNetV2	**80.4 ± 7.4**	**81.3 ± 7.7**	**79.9 ± 8.2**	**79.8 ± 8.1**			**ShuNetV2**	**72.0 ± 1.7**	**72.3 ± 1.6**	**76.7 ± 2.7**	**71.9 ± 1.8**
		EfftV2	74.4 ± 3.4	75.8 ± 3.6	74.3 ± 4.0	73.8 ± 3.6			EfftV2	64.4 ± 3.6	64.6 ± 3.7	66.9 ± 8.0	64.2 ± 3.5
		ViT	70.1 ± 2.8	70.6 ± 3.0	70.3 ± 10.3	69.6 ± 2.6			ViT	63.0 ± 9.5	63.0 ± 9.6	63.0 ± 12.5	63.0 ± 9.6
		PvT-B0	66.3 ± 7.2	67.7 ± 5.2	60.9 ± 15.2	64.0 ± 10.9			PvT-B0	63.0 ± 8.0	58.9 ± 17.4	49.0 ± 28.0	59.1 ± 13.8
		MLP-Mi	67.9 ± 9.0	68.2 ± 9.6	74.8 ± 7.4	67.1 ± 10.1			MLP-Mi	63.0 ± 5.8	64.5± 7.8	62.0 ± 16.7	62.5 ± 5.3
		ResMLP	72.8 ± 5.7	73.3 ± 5.7	75.0 ± 10.9	72.5 ± 5.6			ResMLP	68.5 ± 2.0	69.1 ± 1.8	67.0 ± 7.5	68.2 ± 2.2
	GAF images without HbO Int	ReNet18	70.6 ± 7.2	71.8 ± 8.2	66.9 ± 10.3	69.7 ± 6.8		GAF images without HbO Int	ReNet18	66.5 ± 7.2	67.9 ± 7.9	65.0 ± 4.1	66.0 ± 7.0
		MoNetV2	71.2 ± 3.4	73.0 ± 2.0	70.6 ± 2.8	70.0 ± 4.0			MoNetV2	60.4 ± 1.9	61.0 ± 1.7	61.1 ± 5.3	60.0 ± 2.3
		ShuNetV2	71.2 ± 5.8	71.8 ± 6.2	71.4 ± 6.1	70.9 ± 6.0			ShuNetV2	67.1 ± 3.7	68.0 ± 3.3	56.3 ± 7.3	66.7 ± 4.1
		EfftV2	72.8 ± 5.0	73.4 ± 4.6	73.2 ± 4.7	72.8 ± 4.9			EfftV2	60.0 ± 1.5	60.3 ± 2.1	58.6 ± 4.9	57.5 ± 1.6
		ViT	66.8 ± 3.7	68.3 ± 3.8	63.2 ± 14.1	66.0 ± 3.8			ViT	62.0 ± 1.9	63.8 ± 2.2	66.0 ± 12.4	65.1 ± 2.4
		PvT-B0	64.2 ± 4.5	60.4 ± 16.1	62.8 ± 33.2	58.1 ± 11.9			PvT-B0	56.0 ± 8.5	51.4 ± 15.8	50.0 ± 25.9	51.7 ± 12.5
		MLP-Mi	70.1 ± 6.2	70.2 ± 6.2	64.8 ± 7.0	69.7 ± 6.0			MLP-Mi	58.5 ± 6.2	59.6 ± 6.2	47.0 ± 17.0	56.8 ± 7.5
		ResMLP	70.1 ± 6.7	71.5 ± 7.8	67.3 ± 12.3	69.4 ± 6.2			ResMLP	56.0 ± 4.6	57.5 ± 6.4	44.0 ± 16.6	54.2 ± 4.9
	RP images	ReNet18	69.5 ± 4.2	72.0 ± 7.0	56.8 ± 6.9	68.2 ± 3.2		RP images	ReNet18	61.0 ± 4.4	62.6 ± 5.1	49.0 ± 8.0	60.0 ± 4.2
		MoNetV2	72.8 ± 5.8	73.1 ± 6.2	72.6 ± 5.7	72.5 ± 5.7			MoNetV2	65.4 ± 3.9	65.6 ± 4.1	62.3 ± 4.5	65.3 ± 3.8
		ShuNetV2	67.9 ± 6.3	69.7 ± 5.8	67.7 ± 6.8	66.4 ± 7.5			ShuNetV2	62.4 ± 1.6	65.3 ± 1.5	40.9 ± 3.8	60.6 ± 2.1
		EfftV2	71.2 ± 4.0	71.8 ± 4.4	70.3 ± 3.7	70.3 ± 3.8			EfftV2	62.5 ± 1.5	62.9 ± 1.7	64.0 ± 9.6	62.2 ± 1.6
		ViT	72.3 ± 2.6	74.6 ± 5.0	80.1 ± 14.6	71.4 ± 3.1			ViT	58.0 ± 5.8	58.6 ± 6.2	56.0 ± 14.6	57.3 ± 5.9
		PvT-B0	54.3 ± 3.5	46.9 ± 25.2	41.3 ± 48.0	37.2 ±3.6			PvT-B0	50.5 ± 1.0	40.1 ± 20.2	38.0 ± 44.8	36.3 ± 3.9
		MLP-Mi	69.0 ± 3.1	72.8 ± 6.2	56.8 ± 16.8	67.2 ± 5.1			MLP-Mi	56.0 ± 11.3	56.7 ± 13.1	32.0 ± 21.1	51.8 ± 13.4
		ResMLP	66.3 ± 4.9	73.9 ± 4.9	64.4 ± 26.2	61.6 ± 8.9			ResMLP	57.0 ± 6.0	53.4 ± 15.4	64.0 ± 30.2	51.3 ± 11.4
	MTF images	ReNet18	70.7 ± 3.2	72.8 ± 3.8	61.9 ± 16.4	69.3 ± 2.8		MTF images	ReNet18	63.5 ± 5.4	64.1 ± 5.5	62.0 ± 6.8	63.0 ± 5.9
		MoNetV2	66.3 ± 3.2	67.8 ± 2.6	65.8 ± 3.2	65.0 ± 5.0			MoNetV2	66.8 ± 2.7	67.7 ± 2.4	65.0 ± 3.3	66.3 ± 3.1
		ShuNetV2	69.0 ± 1.4	70.4 ± 2.2	67.8 ± 2.8	67.4 ± 3.6			ShuNetV2	64.9 ± 2.4	65.4 ± 2.8	62.2 ± 9.9	64.6 ± 2.2
		EfftV2	70.1 ± 6.5	70.7 ± 7.3	69.5 ± 6.3	69.4 ± 6.3			EfftV2	65.4 ± 1.9	65.4 ± 1.9	66.9 ± 2.5	65.3 ± 1.8
		ViT	67.9 ± 2.7	69.0 ± 2.4	71.4 ± 13.8	66.7 ± 3.0			ViT	60.5 ± 6.8	61.3 ± 6.5	63.0 ± 14.4	59.0 ± 8.3
		PvT-B0	62.5 ± 3.4	70.6 ± 6.5	55.4 ± 29.7	57.3 ± 9.5			PvT-B0	57.5 ± 10.0	52.9 ± 16.9	52.0 ± 33.3	51.3 ± 15.4
		MLP-Mi	67.4 ± 6.8	68.6 ± 5.5	61.4 ± 25.1	64.2 ± 11.0			MLP-Mi	52.0 ± 6.2	48.2 ± 13.8	38.0 ± 24.2	47.8 ± 9.4
		ResMLP	60.3 ± 2.9	55.8 ± 13.8	64.1 ± 32.3	54.4 ± 9.4			ResMLP	59.5 ± 7.7	56.9 ± 16.5	67.0 ±34.0	53.7 ± 13.1

NOTE: STS-TRANS: SIMPLE TIME SERIES TRANSFORMER; TS-TRANS: TIME SERIES TRANSFORMER; RENET18: RESTNET18; MONETV2: MOBILENETV2; SHUNETV2: SHUFFLENETV2; EFFTV2: EFFTIVENETV2; PVT-B0: PVTV2-B0; MLP-MI: MLP-MIXER; RESMLP: RESMLP.

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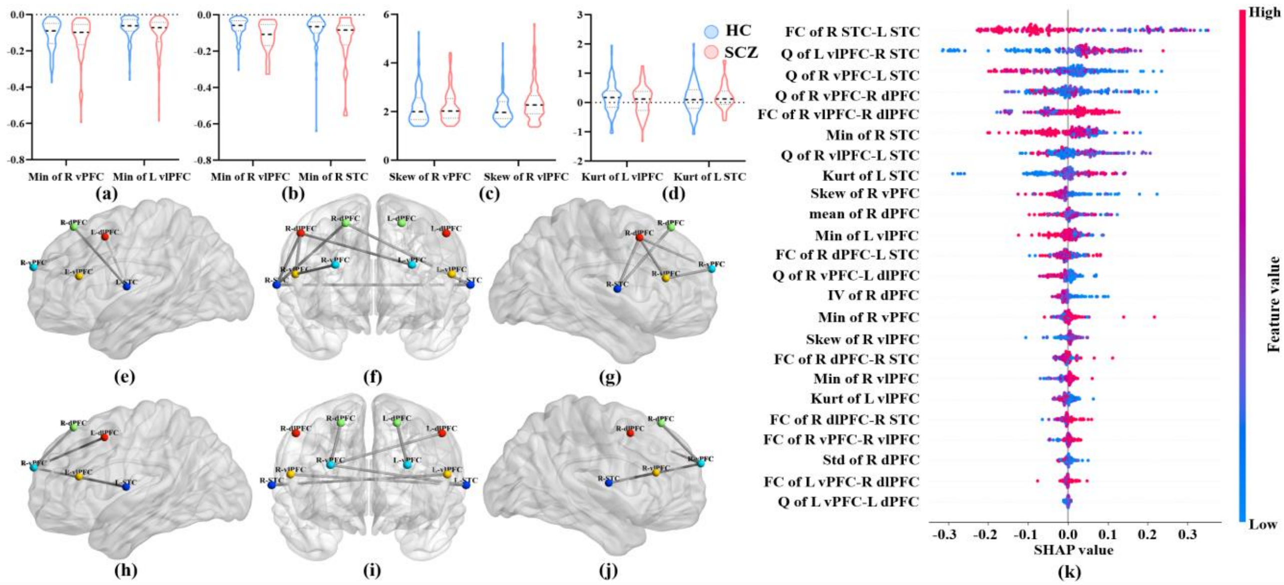

Fig. 7. Features selected through RFECV algorithm; (a-d) Characteristics of HbO activation curve violin diagram, M: mean; (e-g) FC characteristics; (h-j) dFC coefficient variation Q; (k)The SHAP diagram ranks the importance of 24 optimal features, depicting the distribution of their impact on the prediction results along the X -axis. Each point represents a sample, and color represent feature value.

D. Traditional fNIRS Features Selection

Through the RFECV algorithm, 24 optimal features were selected for subsequent classification. As shown in Fig. 7, the violin plot represents the characteristics of HbO concentration changes. Fig. 7(e-g) shows the selected FC features. Fig. 7(h-j) shows the coefficient of variation Q of dFC. The SHAP plot ranked the importance of 24 features, and it can be seen from the SHAP plot, FC, Q, the maximum, kurtosis, and skewness of HbO time series in different brain regions of the HC and SCZ groups mainly affect the machine learning classification results.

E. Classification Between GAF and fNIRS Features

Table IV presents the classification results between the HC and SCZ groups. We evaluated the classification performance of traditional features commonly used in the fNIRS field. All traditional models, including SVM, LDA, RF, LR, XGBoost, and GBDT, achieved accuracies ranging from 64% to 68%. The best-performing traditional model was RF, with an accuracy of 67.6% ± 7.6% and an F1-score of 67.1% ± 9.3%. Overall, the classification performance of conventional fNIRS features (such as HbO activation, FC, and dFC) remains to be fully developed. In contrast, the combination of GAF images with deep learning provides a novel classification approach that clearly outperforms traditional models across multiple evaluation metrics, demonstrating promising potential for SCZ classification.

TABLE IV
CLASSIFICATION RESULTS (MEAN ± STANDARD DEVIATION) OF HC AND SCZ.

Classifier	Accuary(%)	Precision(%)	Recall(%)	F1 (%)
SVM	64.1 ±1.2	64.7 ± 6.4	62.5 ± 12.2	62.9 ± 6.5
LDA	64.7 ± 5.1	64.6 ± 8.3	63.9 ± 6.9	64.1 ± 7.2
RF	67.6 ± 7.6	67.1 ± 9.4	67.1 ± 9.4	67.1 ± 9.3
LR	66.3± 3.0	66.6 ± 5.4	67.0 ± 11.8	66.0 ± 5.0
XGboost	64.7 ± 8.9	64.6 ± 8.1	64.2 ± 12.8	64.1 ± 10.3
GBDT	65.3 ± 6.5	64.7 ± 5.3	66.4 ± 12.4	65.2 ± 8.3

VII. DISCUSSION

In the fNIRS-VFT classification research for SCZ, patients exhibit lower activation intensity. This difference in activation carries important neural activity information and should not be overlooked. Based on this, this study proposes a GAF virtual image encoding method combining activation features. This approach encodes fNIRS signals using GAF with different normalization strategies, preserving changes in HbO concentration and hemodynamic response characteristics. Considering that the handling of different activation curves may affect classification performance, we explored the performance differences between directly using time series, RP, MTF, and GAF images without activation features. Additionally, to comprehensively compare the effectiveness of virtual image and traditional approaches, we also extracted time-domain features, FC, and dFC for classification. Experimental results show that the GAF image combining activation features yield the highest classification accuracy. This method provides a novel analytical perspective for SCZ identification based on fNIRS-VFT.

GAF virtual images are generated from HbO concentration change curves. Its essence is still time-series data. GADF, GASF virtual images contain localized temporal correlations within the time series and correlations between time points before and after. The channel information formed by the light source and detector is also transformed into the channel dimension of the virtual images [31]. GASF and GADF virtual images contain more diverse spatial information than traditional features [31]. At the same time, we noticed that if the intensity of individual channel signals is used for normalization, the generated virtual images may lose the intensity information of different channel concentration variations, which may lead to worse classification results. From the physiological perspective of SCZ disease, the activation intensity of patients during VFT tasks is generally lower than that of HC [3], [15], [20], [21], [24], [37], [61]. For avoiding the loss of activation strength information in GAF virtual images, we used maximum and minimum values of all data for normalizing. Secondly, the changes in pixel values in GAF virtual images are related to time series fluctuations. This fluctuation may be a specific waveform of patients with psychiatric disorders. For example, SCZ patients have been reported to exhibit a more gradual and lower increase in HbO concentration during VFT tasks, and a waveform with inefficient growth reappeared within a period of time after the task ended [37]. Compared to directly using time series fNIRS signals, GAF encoded virtual images can capture subtle changes in activation, improving classification performance.

In virtual image classification experiments, the GAF virtual images achieved the highest classification accuracy under 5-fold cross-validation, outperforming RP and MTF encodings. On the external test set, GAF maintained a relatively superior performance, compared to RP and MTF, demonstrating relatively stable generalizability. Among the three image transformation methods, MTF showed the poorest performance. MTF encodes time series based on first-order Markov transition probabilities. However, during image reconstruction, MTF suffers from ambiguity and inverse uncertainty. It is less sensitive to the amplitude changes typical of physiological signals, resulting in suboptimal classification performance [39]. In contrast, RP constructs a symmetric similarity matrix by computing the pairwise distances (euclidean distance) between time points, emphasizing similarity rather than temporal order [38]. As a result, RP images tend to weaken the representation of temporal trends and amplitude variations, which limits their discriminative power for dynamically varying fNIRS signals. GAF uses the cosine (or sine) of the angular difference or sum between time points to construct the GAF matrix [32]. Compared to other virtual image transformation methods, traditional features that rely on statistical dimensionality reduction, and single-channel time-series signal convolution-based feature extraction approaches, this approach better captures local changes in hemoglobin concentration and preserves the temporal evolution of the signal with relatively low computational cost, which facilitate extraction of both local and global features [34]. Finally, the comparison results of GAF virtual image classification under different normalization strategies show that fusing activation strength information helps to further improve the accuracy of SCZ recognition.

In addition, in this study, virtual images performed better when used together with CNN architecture models. Some fNIRS studies on SCZ have shown that fusing relevant information between different channels can help improve diagnostic performance [3], [17]. In order to further aggregate the relevant information of different brain regions or channels, ShuffleNetV2 is used to rearrange the channel grouping, which helps the model to fuse the relevant information of all channels in the shallow part during the training process. We believe that early recognition of the correlation between channels enhances the model's ability to perceive disease-specific information, which may improve recognition performance.

Although classification using traditional features is less effective than GAF coding. Traditional features still play an important role in the classification of HC and SCZ and can help clinicians further analyze the patient's condition [19]. In this study, the features selected for classification using the RFECV algorithm were mostly FC and the coefficient variation of dFC except for some time-domain features. In the HC and SCZ groups, there were differences in FC between L STC-R STC, R dPFC-L STC, R vlPFC-R dlPFC, and R dPFC-R STC. The neurophysiological function of the left and right vlPFC is related to the symptoms of SCZ. Working memory or complex social behavior is related to right vlPFC activity [62]. Lower FC intensity between STC may be associated with abnormal auditory abilities in SCZ patients [63], [64], [65]. Consistent with the results of other studies, FC dysfunctions and lower activation in the cerebral cortex result in reduced brain area coordination and poor task performance among SCZ patients during VFT tasks [2], [17], [66], [67]. One approach for future work is to integrate GAF virtual images with traditional features to further improve the recognition of disease manifestations.

Even though our study is encouraging, there are some limitations to consider in our study. First, we used clinical data in this research, and patients may have been seen for the first time or reviewed.However, we were unable to obtain information about the patients' previous medication. Some antipsychotic drugs, such as aripiprazole [68] and haloperidol [69], have been reported to influence cognitive function by improving regional cerebral blood flow [70], [71]. These changes in regional barin cerebral blood flow include variations in task-sensitive HbO concentration which is used for subsequent classification analysis. Changes in cerebral hemodynamics may lead to discrepancies between the disease recognition model and the actual clinical state. However, the influence of medication on fNIRS measurement cannot be ruled out in our study. Future studies will require more rigorous experimental designs to control for the influence of various medications and related therapies on participants. Secondly, patients with mental disorders show different degrees of brain function impairment at different clinical stages. Our study focused more on the effect of classification without considering the severity of patients with mental disorders, such as distinguishing between patients with first-episode SCZ or hospitalized patients with SCZ. In future research, we will conduct more detailed grouping of the severity of the disease. Third, this was a single-center study that included only the Mental Health Center of West China Hospital in Sichuan. We tried to control the sample size, age, gender, and data quality as much as possible before conducting subsequent classification analysis, which resulted in a decrease in the sample size. The exclusion of data may affect the generalizability of our research results. Increasing the sample size while ensuring a balanced sample size among different groups, developing conventional models for HC and disease patients based on a large amount of data and enhancing the robustness and generalization of classification performance are our next research aim. Fourthly, considering HbO more directly reflects task-related cortical activation, we only used the concentration changes of HbO for subsequent classification analysis in this study. Nonetheless, HbR and HbT have also been reported in SCZ studies. For example, delayed HbR responses in the frontotemporal regions of SCZ patients have been reported in cognitive tasks [72]. The changes in these bioactive proteins may further enrich the accuracy of SCZ intelligent reality. Therefore, we conducted preliminary testing on the classification results of GAF virtual images converted from HbR and HbT time series, and summarized them in Supplementary Material Part 3. The fusion classification effect of multiple signal virtual images needs further exploration in the future. Finally, specialized classification model was not designed for the fNIRS GAF virtual image encoding framework in our study. We used some simple models and modified some hyperparameters. The performance of other classification models may be underestimated. Our research aims to demonstrate the proposed the strategy of GAF virtual image encoding with activation information is applicable for the recognition of SCZ, and can be extended to assist in the diagnosis and screening of other psychiatric disorders in the future, rather than a specific model architecture.

VIII. Conclusion And Future Work

With the economic development of society, the incidence of SCZ is increasing year by year. It is important to utilize neuroimaging for the screening and classification of disorders. This article proposes a fNIRS-VFT SCZ classification strategy integrating activation information GAF images. The strategy uses modified GAF encoding technique to transform fNIRS signals with different levels of activation into multi-channel images. Through this method, traditional fNIRS sequence classification change to an image classification problem effectively. We collected and used fNIRS data from 200 subjects to demonstrate the effectiveness of this analytical classification method. In addition, we also compared the classification performance fNIRS traditional features, fNIRS time series and other images encoding method. Among these comparison methods, the GAF virtual image with fused activation features performs the best. We hope to improve fNIRS classification effect by introducing deep learning classification research and provide inspiration for other work.

At present, this study is still in an early exploratory stage. In future work, we will further expand the dataset to reduce the potential risk of activation loss caused by global min-max normalization during practical application. Additionally, we plan to combine the GAF encoding method with hemodynamic response features to improve interpretability. Finally, in clinical applications, the classification probability is fed back to psychologists to assist in making diagnostic decisions.

IX. Code and Data Availability

Due to the involvement of personal privacy in the original fNIRS data, we are currently unable to make the full dataset publicly available. However, interested researchers may contact Dr. Yajing Meng at the Mental Health Center, West China Hospital of Sichuan University (yajingmeng218@163.com) to obtain more information regarding participant data collection and usage in this study. In addition, for further details regarding the code of this reserch (including fNIRS preprocessing, GAF virtual image conversion, and classification code), please contact Professor Daifa Wang (daifa.wang@buaa.edu.cn).

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